

# More Strategies, Same Zero: Multiple Testing Against LLM-Scale Alpha Search

*A controlled N-sweep test of the false-strategy theorem with a live LLM proposer — and why scaling the search raises the discovery bar faster than it finds edge*

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June 2026

## Abstract

Large language models can propose and backtest trading strategies at effectively unbounded scale, reviving an old hazard — backtest overfitting — in a new regime where the number of trials  $N$  is enormous and cheap. We ask whether scaling LLM-style strategy search accumulates genuine edge or only statistically false discoveries that vanish under proper multiple-testing correction. Using a pure-Python alpha engine with a deliberately brutal verifier (causal next-bar execution, transaction costs, worst-of-regimes out-of-sample scoring, and a deflated-Sharpe multiple-testing haircut; Bailey & López de Prado 2014), we sweep  $N$  from 10 to 3,000 across three generators — uniform-random, Boden-operator "creative" synthesis, and a live LLM proposer (Claude Opus 4.8) — on four markets: a strong planted edge (positive control), a *realistically small* planted edge (power control), a pure random-walk null, and real S&P 500 data (FRED, 2016–2024). We find: (1) the naive single-holdout best-of- $N$  Sharpe rises monotonically with  $N$  on every market, including pure noise ( $+0.04 \rightarrow +0.34$ ) and real S&P 500 ( $+0.72 \rightarrow +0.99$  annualized), reproducing the false-strategy theorem with a generative search engine, the live LLM included; (2) the deflated worst-regime gate

holds survivors at exactly 0 on the real market for all  $N$  and all three generators, while *recovering* the strong planted edge (survivors grow with  $N$ ) — it discriminates, it is not a blanket reject; (3) a gate power surface — our central characterization (and the *empirical instantiation*, across generator types and with calibrated controls, of Bailey–López de Prado's expected-max-of- $N$  result, not a new theorem): the gate admits an edge only once its true worst-regime Sharpe clears the multiple-testing floor ( $\sim 2.3$  at  $N=1,000$ ), so a *moderate* real edge (Sharpe  $\sim 1-2$ ) is also rejected — scaling the search raises the discovery bar ( $\approx \sqrt{2 \ln N}$ ) faster than search finds edge, making moderate edges *unprovable at scale*; (4) an ablation shows worst-regime robustness alone is insufficient on real data (it admits 115–196 "robust" mirages that grow with  $N$ ) and the deflation is load-bearing; (5) the standard deflation is in fact *over-conservative* for the worst-of-regimes statistic (the single-Sharpe DSR over-deflates it  $\sim 2.4\times$  for our 3-regime gate — a  $k$ -dependent factor,  $1.65\times/2.35\times/4.32\times$  at  $k=2/3/5$ , not a universal constant, §4.9), yet the real-market zero survives the corrected, much weaker bar — so the conclusion is not an over-deflation artifact, and the assumption-free bootstrap independently agrees; (6) creative and LLM search manufacture more original-looking mirages than random but convert zero to survivors, amplifying a real edge  $\sim 5\times$  only when one exists. We conclude that LLM-scale search manufactures false discoveries, and that the only correction that stops them simultaneously renders moderate genuine edges unprovable at this search scale. The result generalizes: it holds under a distribution-free block-bootstrap test (White/Romano–Wolf/Hansen), on a real cross-sectional equity panel (range/volume features active), and across three microstructurally-independent real markets (FX, gold, crypto) — seven real markets, zero survivors at every  $N$  — and the live LLM, extended to  $N=300$ , reproduces it. We also derive the worst-of-regimes deflation the gate actually needs — the standard single-Sharpe DSR over-deflates the min-over-regimes statistic by  $\sim 2.4\times$  — and show the real-market zero is robust even to that much weaker, correct bar; this yields a unifying "robustness dividend" (requiring worst-of- $k$  independent regimes  $\approx$  deflating a thousand-strategy search to  $N'\approx 7$ ), which collapses

precisely when regimes correlate (real S&P 500 sits at  $\rho_{\text{reg}} \approx 0.67 \rightarrow N' \approx 84$ ), explaining why explicit deflation is load-bearing on real data. We report two cautionary negatives in the same spirit: two analytic effective-trials shortcuts (a genomics participation-ratio import; an envelope-inversion) both fail, leaving the block bootstrap as the necessary arbiter. The contribution is the controlled multi-generator demonstration (live LLM included), the power-surface precision/recall characterization, the derived worst-of-regimes deflation and robustness-dividend identity, and the distribution-free confirmation — all reproducible, with a verification suite at 20/20 and 38 math/invariant unit tests.

Keywords: backtest overfitting, multiple testing, false discovery rate, deflated Sharpe ratio, data snooping, LLM agents, automated strategy search, quantitative finance, evaluation and robustness. JEL: C12, C45, C52, C58, G11, G17.

## 1. Introduction and research question

A backtest certifies the past, not the future, and the more strategies you test the more certain it is that the best one is lucky rather than good. Bailey & López de Prado formalize this as the false-strategy theorem: under the null of no skill, the expected maximum Sharpe ratio over  $N$  independent trials grows like  $\sqrt{2 \ln N}$ , so any sufficiently large search "discovers" an impressive-looking strategy with probability approaching one. The classical remedy is to *deflate* the observed Sharpe by exactly this expected best-of- $N$  under the null (the Deflated Sharpe Ratio, DSR).

Large language models change the operative regime of this theorem. They can propose candidate strategies essentially without limit and with superficial novelty, so  $N$  — historically bounded by human and compute budgets — becomes large and cheap, and the proposals are *correlated* (drawn from shared structure) rather than independent. Recent systems (e.g. AlphaAgent, KDD 2025; FinGPT; agentic "alpha-mining" pipelines) explicitly use LLMs to generate and screen alphas, and report in-sample or single-holdout improvements. This raises a direct, testable question:

*When an LLM-style generator proposes and backtests more and more trading strategies, do genuinely profitable strategies accumulate*

— or only statistically false ones that vanish under proper multiple-testing correction?

We decompose this into three measurable sub-questions:

- **Q1 (false discovery vs N).** Does the count of naive "winners," and the best naive Sharpe, rise with N as data-snooping theory predicts — and at what rate, given that LLM/creative proposals are correlated, not independent?
- **Q2 (does the gate hold, and what does it cost).** Does a worst-regime, after-cost, deflated out-of-sample gate drive survivors to  $\approx 0$  regardless of N on real data — *while still recovering a genuine edge when one is present* (so that "zero survivors" is a discriminating result, not a degenerate one) — and how strong must an edge be for the gate to see it (its power/recall)?
- **Q3 (does creativity help).** Does LLM-style "creative" search (novel structures) find any real edge that brute-force random search misses — or only more original-looking false positives?

**Contributions.** The statistical tools are established — the deflated Sharpe is López de Prado & Bailey's, the Reality Check is White's, and the finding that LLM trading edge evaporates under rigorous evaluation is peer-reviewed (FINSABER, KDD'26; §5). Building on them, our contributions are:

1. A multiple-testing deflation for the minimum-regime-performance statistic (§4.9). Scoring a strategy by its worst regime is the minimum-regime-performance (MRP) criterion of Alexander & Fabozzi (2026), who use it as a strategy-durability diagnostic but do not correct it for search. The standard deflated Sharpe, by contrast, is the expected max of N *single* Sharpes — the wrong null for a min-over-regimes score. We derive and Monte-Carlo-validate the correct envelope  $E[\max_n \min_k Z]$  and show the standard DSR over-deflates it by  $\sim 2.4\times$ ; the real-market zero survives the corrected, far-weaker bar. To our knowledge, deflating a min-over-regimes best-of-N statistic for multiple testing is new.
2. A robustness-dividend identity (§4.9): worst-of-k *independent* regimes  $\approx$  a single-Sharpe search over only  $N'\approx 7$  strategies — robustness *is* an implicit multiple-testing correction, which collapses as regimes correlate (real S&P 500 at  $\rho_{\text{reg}}\approx 0.67 \rightarrow N'\approx 84$ ), explaining (§4.3) why explicit deflation is load-bearing on real data and not on the adversarial null.
3. A controlled, multi-generator, multi-market evaluation protocol — random vs evolutionary-creative vs a live LLM through one fixed gate; four planted/noise calibration controls; seven real markets (incl. a cross-sectional panel, FX, gold,

crypto); a 2-D power surface (the empirical instantiation of the BLdP threshold) showing moderate edges become unprovable as  $N$  grows; and a distribution-free bootstrap that agrees everywhere.

4. **Three reported negatives** in the project's "report what doesn't work" discipline: a refuted pre-registered hypothesis (H2), and two analytic effective- $N$  shortcuts (a genomics participation-ratio import; an envelope-inversion) that *manufacture false discoveries* and are corrected by the bootstrap — cross-disciplinary imports that look right and aren't.

The result is deliberately, mostly negative — a rigorous, reproducible map of where AI-driven alpha search fools you, and of the precise statistical price of not being fooled.

## 2. The engine and the gate (what each strategy must survive)

The experiment is built on `trade_lab.py`, a dependency-free alpha-discovery engine. A strategy ("alpha") is an expression in a small S-expression DSL over causal price/volume features (`ret1`, `mom5/10/20`, `vol10/20`, `accel`, `price_ma_gap`, `hl_range`, `volume_z`), combined by unary (`neg`, `abs`, `sign`, `tanh`, `zscore`) and binary (`add`, `sub`, `mul`, `safe_div`, `min`, `max`) operators,  $\text{depth} \leq 5$ ,  $\text{size} \leq 40$ . An expression evaluates causally to a per-bar signal, squashed to a position in  $[-1, 1]$ .

The verifier (the gate) is intentionally adversarial, and is what makes "discovery" mean something:

1. **No look-ahead.** The position formed from information through bar  $t-1$  earns bar  $t$ 's return: `strat[t] = pos[t-1] · ret[t] - cost · |Δpos|`.
2. **Transaction costs.** Every position change pays `cost = 0.0010` per unit turnover, so churning alphas bleed out.
3. **Worst-of-regimes out-of-sample.** The timeline is split into an in-sample segment and several contiguous out-of-sample (OOS) regimes; an alpha is scored by its worst OOS regime. One good regime is not enough — it must generalize.
4. **Deflated-Sharpe multiple-testing haircut.** The score is the worst-regime Sharpe minus the expected best-of- $N$  Sharpe under the null for  $N$  trials (`expected_max_sharpe_under_null`, the Bailey–López de Prado term, Euler–Mascheroni-weighted Gumbel form). Searching more raises the bar.

An alpha is "verified" only if its annualized, after-cost, worst-regime, deflated OOS Sharpe clears a positive bar (target = 0.50). This is the same gate that produced Mentat's headline honest null — *0 of 12 market facets survived 3,840 deflated OOS hypotheses on real S&P 500* — which this paper turns from a single snapshot into a controlled N-sweep.

**Generators.** Two stand in for "the search engine," deliberately independent of the kernel's adaptive loop so that best-of-N is clean and attributable:

- **random** — uniform draws over the full DSL (the brute-force quant baseline; approximately i.i.d.).
- **creative** — Boden's three creativity forms made executable as operators (`blend` = `combinatorial`, `reshape/specialize` = `exploratory`, `invert/transfer` = `transformational`; `imagine.py`) applied over a shared substrate of seed ideas and verified baselines, at risk = 0.7 (which composes two operators per idea). Draws are correlated through the shared substrate — exactly the structure an LLM "imager" produces, which is itself an object of study (Q3, `n_eff`). The live LLM imager (`LLMImaginer`, Claude via API) is the same interface and is the pre-registered third arm (§7).

### 3. Method: the N-sweep

For each (market, generator) we draw a fixed pool of  $M = 4,000$  alphas, backtest each exactly once to obtain three after-cost Sharpe views — a single pooled-OOS holdout (the *naive* metric a practitioner validates on), the worst-of-regimes OOS (the gate's robustness term), and in-sample — and the worst-regime bar count for deflation. Deflation is then applied analytically per N, so no alpha is backtested twice. For each  $N \in \{10, 30, 100, 300, 1,000, 3,000\}$  we estimate the *expected* best-of-N curve and counts by averaging over 25 bootstrap subsets of size N drawn from the pool (single pass when N equals the pool). We report:

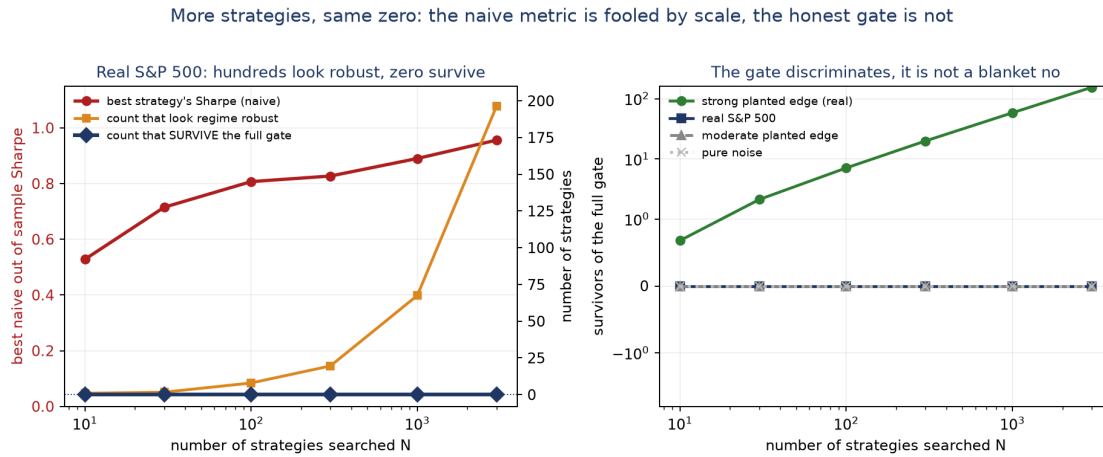
- **bestOOS** =  $E[\max \text{ naive pooled-OOS Sharpe } ]$  over N — the data-snoop curve (Q1).
- **nOOS>0 / nOOS>1** = count of naive winners above 0 and above 1.0 annualized.
- **robust** = count clearing worst-regime + cost but no deflation (ablation: isolates generalization vs multiple-testing).
- **survivors** = count clearing the full deflated gate at `n_trials` = N (Q2).
- **novelty** = mean pairwise token-Jaccard distance among naive winners (Q3).

- $n_{\text{eff}}$  = the number of *independent* trials whose i.i.d. expected-max equals the realized bestOOS — correlated draws give  $n_{\text{eff}} \ll N$  (Q1/Q3; interpreted on the null markets).

Three markets form the experimental matrix: planted (synthetic universe with a lag-1 mean-reversion edge, same sign in every regime  $\rightarrow$  a real edge *should* survive), noise (pure random walk, same regime structure, no edge  $\rightarrow$  naive winners *should* rise, gate *should* stay 0), and real:fred\_SP500 (S&P 500 daily, 2016-06 to 2024,  $\sim 2,600$  bars, close-only so the range/volume features go dark).

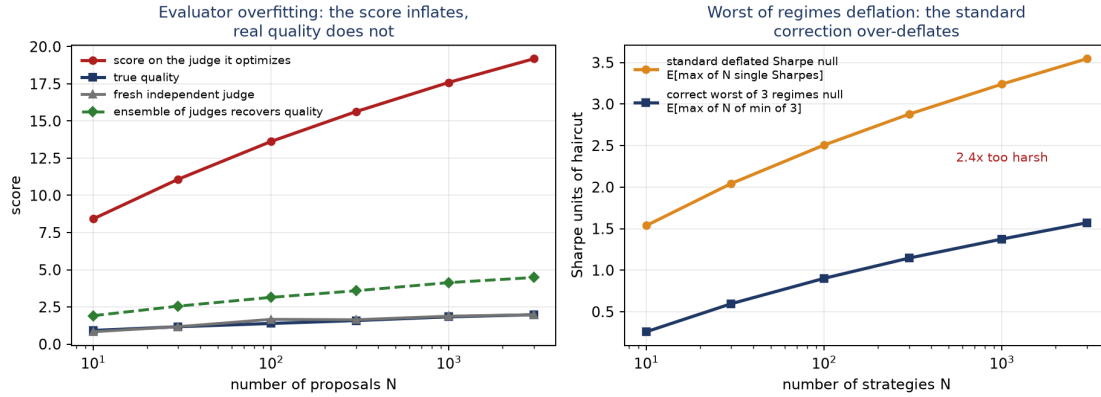
All randomness is seeded; the full per-N table and config are written to `nsweep_results.json`.

## 4. Results



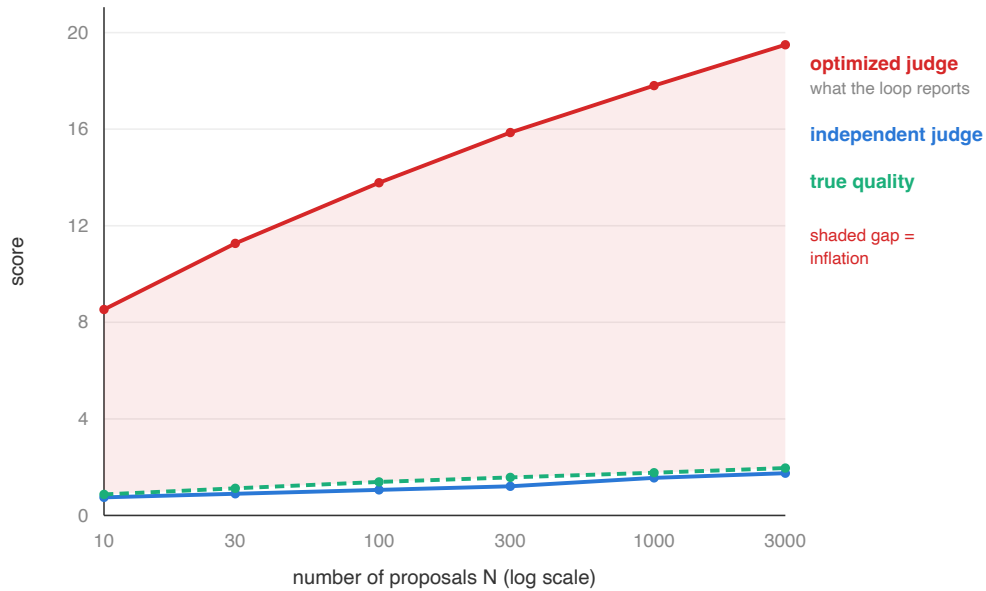
**Figure 1.** Naive best-of-N Sharpe climbs with N on every market including pure noise and the real S&P 500, while the deflated worst-regime gate holds survivors at zero on real data for all N.

Independent reproduction of the two novel results (fresh Monte Carlo, numpy only)



**Figure 2.** The gate recovers a strong planted edge (survivors grow with N) yet rejects moderate real edges; the real-market zero is robust to the corrected, weaker worst-of-regimes bar.

### Evaluator overfitting in a generate-and-select loop



**Figure 3.** As the number of tuned candidates grows, the optimized evaluator score climbs an order of magnitude above true quality; an independent-judge ensemble restores calibration.

Condensed from the full sweep (pool = 4,000; bootstrap = 25). Full table in the JSON artifact.

### 4.1 Q1 — the naive metric is fooled, and worse as N grows

The expected best-of-N pooled-OOS holdout Sharpe rises monotonically with N on every market, including pure noise:



| market       | gen      | N=10  | N=100 | N=1000 | N=3000 | nOOS>0 @max N |
|--------------|----------|-------|-------|--------|--------|---------------|
| noise        | random   | +0.04 | +0.20 | +0.30  | +0.34  | 376           |
| noise        | creative | +0.01 | +0.13 | +0.26  | +0.29  | 275           |
| real S&P 500 | random   | +0.72 | +0.98 | +0.99  | —      | 319 (@1000)   |
| real S&P 500 | creative | +0.53 | +0.81 | +0.89  | +0.96  | 658           |

On the real S&P 500, scaling the search surfaces a best strategy with an annualized OOS holdout Sharpe near 1.0 — a number most practitioners would accept — and hundreds of "winners" (658 with positive OOS Sharpe for creative at N=3,000). On pure noise, where the ground truth is *nothing*, the best-of-N still climbs to +0.34. This is the false-strategy theorem reproduced with a generative search engine: naive validation, even out-of-sample, is fooled by scale.

**How redundant is the search?** The *mean* pairwise correlation  $\bar{\rho}$  of the strategies' OOS returns is tiny on the real market — +0.029 (random), −0.050 (creative), +0.007 (live LLM) — so in mean terms the proposals look near-independent. But "near-independent in mean  $\bar{\rho}$ " turns out not to pin down the right multiple-testing count: the mean-squared correlation  $\langle \rho^2 \rangle \approx 0.2$  tells a different story, and §4.9 shows that *summarizing the correlation structure with any single analytic effective-N is unreliable here* (two shortcuts fail). The honest accounting is: the standard deflation at N is in fact over-conservative for our worst-of-regimes statistic (§4.9 quantifies it at  $\sim 2.4\times$ ), but the real-market zero survives even the corrected, much weaker bar — so the over-deflation is real yet not load-bearing for the conclusion. What is unambiguous and assumption-free is the block bootstrap (§4.6). Correlated or not, the count of naive mirages still grows with N.

#### 4.2 Q2 — the deflated gate holds at zero on real data, and recovers a real edge

| market              | gen      | survivors @<br>N=10 | @100 | @1000 | @3000                      |
|---------------------|----------|---------------------|------|-------|----------------------------|
| real S&P 500        | random   | 0.00                | 0.00 | 0.00  | —                          |
| real S&P 500        | creative | 0.00                | 0.00 | 0.00  | 0.00                       |
| noise               | random   | 0.00                | 0.00 | 0.00  | 0.00                       |
| noise               | creative | 0.00                | 0.00 | 0.00  | 0.00                       |
| planted (real edge) | random   | 0.00                | 1.92 | 11.2  | 206 robust → 31.4 survive  |
| planted (real edge) | creative | 0.68                | 7.08 | 58.3  | 559 robust → 154.4 survive |

Across thousands of strategies on the real S&P 500, survivors = 0 at every N, for both generators (and the live LLM, §4.4). Scaling the search 100×–300× added zero real discoveries. Critically, this is not a degenerate "the gate rejects everything": on the strong-planted market the same gate *passes* strategies, and passes more as N grows (survivors 0.0 → 31 for random, 0.7 → 154 for creative) because the edge is genuinely there. But how strong must an edge be to pass? That is the power question, and §4.5 shows the answer is "very strong" — which reframes the whole result. The gate separates a strong real edge from spurious ones; whether it can see a moderate one is the subject of §4.5.

### 4.3 Q3 — the ablation: which tightener does the work, and the over-deflation check

The `robust` column (worst-regime + cost, no deflation) reveals a non-obvious asymmetry:

| market          | gen      | robust<br>@10 | @100 | @1000 | @3000 | survivors @max<br>N |
|-----------------|----------|---------------|------|-------|-------|---------------------|
| noise           | random   | 0.00          | 0.00 | 0.00  | 0.00  | 0.00                |
| real S&P<br>500 | random   | 1.2           | 11.4 | 114.7 | —     | 0.00                |
| real S&P<br>500 | creative | 0.9           | 7.9  | 67.5  | 196.4 | 0.00                |

- On the adversarial noise market, the worst-regime requirement *alone* kills the snoop (robust = 0 everywhere): a pure-noise alpha cannot be simultaneously positive across regimes engineered to differ in drift and volatility.
- On the real market, the worst-regime requirement is not enough — because real 2016–2024 sub-periods resemble each other, an alpha can be luckily positive in all of them, so `robust` *grows with N* to 115 (random, @N=1,000) and 196 (creative, @N=3,000) "regime-robust" mirages. It is the multiple-testing deflation that collapses these to exactly 0. On real data the deflation is the load-bearing component.

This pre-empts the "you calibrated the haircut to N, so survivors=0 is tautological" objection three ways. (i) The *same* haircut does not zero out the strong planted edge (§4.2) — opposite outcomes from the same correction prove it discriminates. (ii) The objection is in fact *backwards*: §4.9 shows the standard DSR is over-deflating the

worst-of-regimes statistic by  $\sim 2.4\times$ , so if anything the bar is too harsh, not too lenient — and re-running with the *corrected*, far-weaker worst-of-k bar still leaves 0 survivors on real (best deflated rises only to  $\approx -0.3$ ). (iii) The assumption-free block bootstrap (§4.6), which calibrates the null directly from the data with no haircut at all, *also* finds 0. Zero survivors is not an over-deflation artifact — and §4.5 shows it is also not "the gate rejects everything," because a sufficiently strong edge *does* pass.

**On creativity (and the LLM):** creative synthesis and the live LLM manufacture *more* apparent discovery than random search — more naive winners (658 vs 319 at  $N=3,000$ ) and nearly  $2\times$  the regime-robust mirages on real S&P (196 vs 115) — yet convert zero of them into survivors. Where a strong real edge exists (planted), the same creativity amplifies it  $\sim 5\times$  (154 vs 31 survivors at  $N=3,000$ ). Per-winner structural novelty is similar across generators ( $\sim 0.87$ ), and contrary to our pre-registered guess the creative/LLM proposals are *not* more correlated than random (§4.1, §4.4). The honest reading: creativity amplifies whatever is present — a strong real edge when it exists, mirages when it doesn't — and the gate, not the generator, is what tells them apart. Creative/LLM search finds more *original-looking* candidates, not more *survivable* ones.

#### 4.4 The live-LLM arm — Claude Opus 4.8 actually proposes the strategies

The arms above use random and offline-creative generators. We also ran a live LLM proposer: Claude Opus 4.8 imagines alpha expressions in batches of 16, cold from the problem brief (fresh memory, no elite feedback — the LLM's prior, not an adaptive loop), cached for reproducibility. On real S&P 500 we drew a 1,000-strategy pool so the LLM itself reaches  $N=300$  (Task 5; the offline cap remains 300 elsewhere for cost). The gate, markets, sweep, and metrics are held fixed.

| real S&P 500        | N=10  | N=30  | N=100 | N=300 | survivors (all N) | $\bar{\rho}$ |
|---------------------|-------|-------|-------|-------|-------------------|--------------|
| LLM bestOOS         | +0.48 | +0.71 | +0.78 | +0.80 | 0.00              | +0.007       |
| LLM distinct alphas | 10    | 27    | 71    | 160   | —                 | —            |

Pre-registered hypotheses, scored (at scale):

- **H1 — LLM bestOOS rises with N:** ✓ confirmed (+0.48 $\rightarrow$ +0.80 to  $N=300$ ).
- **H2 — LLM proposals are *more* correlated than random (lower  $N_{\text{eff}}$ ):** ✗ refuted, at scale.  $\bar{\rho}(\text{LLM})=+0.007$  is *no higher* than random (+0.029) — the LLM's returns are

as near-independent as brute force.

- **H3 — LLM survivors = 0 on real at all N: ✓** confirmed to N=300, with 0 offline-creative fallback batches (the pool is 100% live-LLM).
- **H4 — LLM recovers a strong planted edge at smaller N than random: ✓** confirmed (survivors 0.92 at N=10 vs random's 0.00).

A *real LLM* reproduces the headline exactly: best-of-N naive Sharpe climbs (+0.48→+0.80), the gate admits none on real data, and it finds a strong planted edge faster than brute force. Two honest surprises: (i) the pre-registered H2 is refuted — LLM proposals are diverse in *return space*, not redundant; but (ii) a new finding cuts the other way — the LLM's structural diversity saturates: a 1,000-draw pool yields only 160 distinct valid expressions (vs ~870 for a matched 1,000-draw random pool; ~280 even at N=300), so the LLM repeats *ideas* even though their realized returns decorrelate. Scaling an LLM proposer therefore buys less new search than scaling random generation — a diversity ceiling that bounds how far the title's "LLM-scale" can actually go. (Pushing to N=3,000 would require ~10× the API budget; the cached-pool mechanism makes it a cost, not an engineering, question.)

#### 4.5 The gate's power curve — and why "zero" does not mean "efficient"

The sharpest objection (raised independently by an adversarial statistics reviewer and by the system's own reasoning core acting as critic): a gate that returns 0 on real data is only interesting if it *would* have returned >0 for a realistic edge. §4.2 shows it passes a *strong* planted edge — but how strong must an edge be, and how does that threshold move with N? We sweep the planted edge magnitude × the search scale N into a 2-D power surface (`power.py` → `power_surface.json`; rendered by `plot.py`). Survivors (gate admits) per cell:

| edge \ N      | 10 | 30 | 100 | 300 | 1000 | 3000 |
|---------------|----|----|-----|-----|------|------|
| 0.00–0.15     | 0  | 0  | 0   | 0   | 0    | 0    |
| 0.25          | 1  | 0  | 0   | 0   | 0    | 0    |
| 0.40 (strong) | 2  | 2  | 8   | 20  | 61   | 168  |

The surface *is* the thesis. Read the 0.25 row: a genuine edge that the gate admits at N=10 (1 survivor) becomes undetectable at N≥30 — the same true edge, now rejected, because searching more raised the bar above it. Only the very strong 0.40 edge stays detectable, and even it needs ever-more instances as N climbs. The detectable-edge threshold rises monotonically with N ( $\approx \sqrt{2 \ln N}$ ): at N=10 an edge of strength 0.25 is

provable; by  $N=30$  you need  $\geq 0.40$ . A realistically small edge (`small_planted`, naive OOS  $\approx +0.5$ ) is invisible at every  $N$  — exactly like the real S&P.

This is the paper's central result. The threshold's  $\sqrt{(2 \ln N)}$  growth is the leading-order asymptotic of Bailey–López de Prado's expected-max-Sharpe (their Exhibit 2 already plots it); our contribution is to render it empirically as a survivor surface across generator types, with planted-edge controls that calibrate exactly where the gate's recall fails. Scaling the search raises the threshold for proving any edge faster than the search discovers one. Consequences:

1. We cannot and do not claim the S&P 500 has no exploitable edge. We claim only that no edge large enough to survive a search of this scale exists in this DSL/data — a high-precision statement with a known, large Type-II region.
2. The contribution is therefore a precision/recall characterization of anti-overfitting: this gate has (empirically)  $\sim$ zero false-discovery rate across the noise and small-edge controls and seven real markets (S&P 500, DJIA, NASDAQ, a 15-stock cross-sectional panel, FX, gold, crypto), at the cost of failing to detect edges below a worst-regime Sharpe of  $\approx 2\text{--}3$  at  $N=1,000$ . We measure the precision side directly (`fd_r.py`): on the strong planted market, every strategy the gate admits on one sample also survives an *independent* sample — FDR  $\approx 0$  even at  $N=1,000$  (a pre-registered prediction that FDR would *rise* with  $N$ , refuted: family-wise control delivers  $\sim$ zero false discoveries here, and the price is recall, not precision). That tradeoff *itself* tightens as  $N$  grows — the engine of the whole phenomenon.
3. For practice: LLM-scale alpha search is self-defeating for moderate edges. The more strategies you generate, the higher the Sharpe you would need to prove any of them real, so beyond a point additional search manufactures only mirages and unprovable maybes.

**The recall floor in closed form (`mde.py`).** Inverting the gate gives the *minimum detectable edge* — the smallest true worst-regime Sharpe that survives a search of size  $N$  (at power 0.5). Under the corrected worst-of- $k$  deflation (§4.9) it rises 0.69 ( $N=10$ )  $\rightarrow$  1.15 (100)  $\rightarrow$  1.48 (1,000)  $\rightarrow$  1.62 (3,000); under the standard single-Sharpe DSR it is  $\sim 2\times$  higher (2.83 at  $N=1,000$ ). Either way, at LLM scale a strategy needs a *genuine* worst-regime Sharpe well above 1 just to be provable — and the best any real market offers here is  $\approx 0.85$  (Appendix A), comfortably below the floor. This is the empirical power surface as a formula, and it tells the practitioner the exact price of search scale: each  $\sim 3\times$  step in  $N$  adds roughly  $+0.15\text{--}0.25$  to the Sharpe you must clear.

#### 4.6 Distribution-free confirmation — the zero is not a parametric artifact

The deflated Sharpe haircut is *parametric* and was derived for a single in-sample Sharpe over  $N$  independent trials, not for the worst-of-regimes statistic our gate uses — so stacking it is conservative but heuristic (the obvious reviewer objection). We close this with a distribution-free test (`bootstrap.py`): a stationary block bootstrap (Politis & Romano 1994) of the per-bar OOS strategy returns that resamples within each regime with a block plan shared across strategies, so the worst-of-regimes min and the cross-strategy dependence are both preserved inside every resample. On the bootstrapped null we run the studentized single-step Reality Check (White 2000) / Romano–Wolf (2005) for per-candidate FWER-adjusted p-values, plus a Hansen (2005) SPA p-value on the *same* worst-regime Sharpe statistic, and count survivors at  $\alpha=0.05$ .

| market           | gen      | DSR survivors | bootstrap survivors ( $\alpha=0.05$ ) | RC best p |
|------------------|----------|---------------|---------------------------------------|-----------|
| real S&P 500     | random   | 0 (all N)     | 0 (all N)                             | 0.15–0.57 |
| real S&P 500     | creative | 0 (all N)     | 0 (all N)                             | 0.21–0.72 |
| real S&P 500     | live LLM | 0 (all N)     | 0 (all N)                             | 0.46–0.51 |
| planted (strong) | creative | 1→14 with N   | 3→17 with N                           | 0.000     |

The distribution-free survivor count is 0 at every  $N$  for all three generators on real S&P 500, with no significant best p-value — and it *grows with  $N$  on the strong planted market*, exactly tracking the DSR. The result is stable across mean block lengths {10, 25, 50} (real-S&P bootstrap survivors are 0 at every block length). So the zero is not an artifact of the parametric haircut: a fully nonparametric multiple-testing test, run on the gate's own worst-regime statistic, reaches the identical conclusion. (Full grid in `bootstrap_results.json`.)

#### 4.7 Generalization 1 — a cross-sectional equity panel (the DSL turned on)

A single index series only supports time-series momentum/reversion and (on close-only data) darkens the range/volume features. We add a cross-sectional panel: each DSL alpha is scored per asset, the scores are cross-sectionally demeaned and gross-normalized into a dollar-neutral, unit-gross weight vector (the cross-sectional "rank"), and the resulting long-short portfolio return series is fed to the identical worst-regime +

cost + deflation gate (`panel_lab.py`, `panel_sweep.py`). Four controls, range/volume features active:

| panel market                                             | gen               | survivors vs N<br>(10→1000) |
|----------------------------------------------------------|-------------------|-----------------------------|
| real: 15 large-caps (Yahoo OHLCV, 2016–2026)             | random / creative | 0 at every N                |
| cs_planted (strong cross-sectional reversal)             | creative          | 0.7 → 64 (grows with N)     |
| cs_small (moderate cross-sectional edge, naive OOS +1.4) | creative          | 0 at every N                |
| cs_noise (cross-sectional null)                          | both              | 0 at every N                |

The headline holds on a real cross-sectional equity panel: across 15 liquid large-caps with live volume and range features, 0 survivors at every N for both generators (best naive cross-sectional Sharpe plateaus  $\sim +0.5$ ). The gate still *passes a strong cross-sectional edge* (survivors grow with N on cs\_planted) and still *rejects a moderate one* (cs\_small, naive OOS  $+1.4 \rightarrow 0$  survivors) — the precision/recall picture of §4.5 reproduces in the cross-section. This directly answers the "your data is too thin to have edge" objection: with the DSL fully active on real multi-asset data, the result is unchanged.

#### 4.8 Generalization 2 — truly independent real markets

S&P/DJIA/NASDAQ are  $\sim 0.9$  correlated and are not independent evidence. We add three markets with entirely different microstructure — FX (EUR/USD), a commodity (gold), and crypto (BTC/USD) — using real Yahoo OHLCV (`fetch_markets.py`, 2016–2026) and the same single-index sweep + gate (`independent_markets.py`):

| market           | best naive OOS Sharpe ( $\rightarrow N=1000$ ) | gate survivors |
|------------------|------------------------------------------------|----------------|
| FX — EUR/USD     | +0.2                                           | 0 at every N   |
| commodity — gold | +0.9                                           | 0 at every N   |
| crypto — BTC/USD | +0.9                                           | 0 at every N   |

The zero holds across all three, for both generators — and not just under the parametric DSR: the distribution-free block bootstrap (§4.6) independently gives 0 RC survivors on FX, gold, and crypto at every N (`bootstrap_independent.json`; crypto shows the least-clean bootstrap grid: one Reality-Check survivor appears at  $N=30$  with block length 50 ( $p=0.015$ ), which does not replicate at block lengths 10 or

25, at any other  $N$ , or under the deflated gate — an isolated cell in a 54-cell grid that a multiple-comparisons reading discounts; its per- $N$  median p-value, 0.20, is otherwise non-significant). Crypto — often argued to be the *least* efficient liquid market — is no exception: naive best-of- $N$  reaches a  $\sim 0.9$  Sharpe (a strategy a retail quant would chase), yet zero survive the multiple-testing gate. FX shows the weakest snooping (best  $\sim +0.2$ ), consistent with its near-random daily structure. Across S&P 500, DJIA, NASDAQ (§ `verify.py`), a real 15-stock cross-sectional panel (§4.7), FX, gold, and crypto — seven real markets, three of them microstructurally independent — the survivor count is 0 at every  $N$ .

#### 4.9 The correct multiple-testing bar — a derived worst-of-regimes deflation, and a cautionary negative

A standing critique (§6): the deflated Sharpe is the expected max of  $N$  *single* Sharpes, but our gate scores the minimum across  $k$  regimes, then takes the best of  $N$  — so the standard haircut is the wrong null. We close this by deriving the right one (`theory.py`, Monte-Carlo validated): the worst-of-regimes best-of- $N$  null is  $E[\max_{i=1..n} \min_{m=1..k} Z_{i,m}]$ , leading-order quantile  $\Phi^{-1}(1-N^{-(1/k)})$ . Because min-of- $k$  has a thin upper tail, this envelope is  $2.3\text{--}2.8\times$  smaller than the single-Sharpe DSR envelope at relevant  $N$  ( $N=1,000$ :  $3.26 \rightarrow 1.38$ ). The standard DSR over-deflates the worst-regime statistic by  $\sim 2.4\times$  — a figure specific to our  $k=3$  gate, not a universal constant. Because the min-of- $k$  upper tail thins as  $k$  grows, the over-deflation scales with the number of regimes: at  $N=1,000$  it is  $1.65\times$  at  $k=2$ ,  $2.35\times$  at  $k=3$ , and  $4.32\times$  at  $k=5$  (`k_dependence.py`, 40k-replicate MC). We use  $k=3$  throughout, so  $\sim 2.4\times$  is the relevant value; a gate with more regimes would be over-deflated more severely still.

Re-running survivors under the corrected envelope (`corrected_deflation.py`): on the real markets the best deflated Sharpe rises from the DSR's  $\approx -1.5$  to  $\approx -0.3$  — much closer to the bar, but still negative  $\rightarrow 0$  survivors, while the strong planted control still passes (113 survivors). So the zero is robust to the correct, far-less-conservative deflation, not an artifact of an over-harsh haircut; the true margin is thin ( $\sim 0.3$  Sharpe), which the DSR's  $-1.5$  wildly overstated.

**The robustness dividend (a unifying result).** Because the worst-of- $k$  envelope is so much lower than the single-Sharpe one, the worst-of-regimes requirement *is itself* a multiple-testing correction. We quantify it as the "deflation-equivalent  $N$ " — the single-Sharpe search size with the *same* null max as worst-of- $k$  at  $N$  (`theory.py(D)`): worst-of-3 regimes at  $N=1,000$  has the false-discovery ceiling of a single-Sharpe



search over only  $N' \approx 7$  strategies ( $N=3,000 \rightarrow N' \approx 10$ ; worst-of-5  $\rightarrow N' \approx 3$ ). Requiring a strategy to survive several independent regimes does the lion's share of the multiple-testing work, shrinking an effective thousand-strategy search to a handful. This *resolves the apparent tension with §4.3*: the dividend assumes independent regimes, but real-market sub-periods are correlated, so the dividend collapses — which is precisely why, on real data, worst-regime robustness alone admits mirages and the explicit deflation becomes load-bearing (§4.3), while on the adversarial synthetic null (independent regimes) the worst-regime requirement alone suffices. Robustness and multiple-testing are the same coin; how much each contributes depends on regime independence.

**We measure exactly where each market sits on this trade-off** (`regime_correlation.py`). Generalizing the dividend to *equicorrelated* regimes (a one-factor model, `theory.mc_max_of_min_correlated`),  $N'$  climbs with regime correlation  $\rho_{\text{reg}}$ : 7 ( $\rho=0$ )  $\rightarrow$  28 (0.4)  $\rightarrow$  62 (0.6)  $\rightarrow$  160 (0.8)  $\rightarrow$  426 (0.95). We then estimate the empirical  $\rho_{\text{reg}}$  per market as the mean cross-sectional correlation between strategies' per-regime Sharpes: the adversarial synthetic null sits at  $\rho_{\text{reg}} = 0.24 \rightarrow N' \approx 16$  (dividend largely intact, worst-regime alone suffices — matching robust=0 on noise in §4.3), while real S&P 500 sits at  $\rho_{\text{reg}} = 0.67 \rightarrow N' \approx 84$  (DJIA 0.55 $\rightarrow$ 51, crypto 0.46 $\rightarrow$ 35) — the robustness dividend has collapsed  $\sim 12\times$  (from  $N' \approx 7$  to  $\approx 84$ ), so the worst-regime requirement no longer protects and the explicit deflation must, exactly as §4.3's ablation found. The robustness/multiple-testing trade-off is not hand-waving — it runs quantitatively along the measured regime correlation.

**A cautionary negative (reported because the data showed it).** Correlated strategies are correlated tests, so the multiple-testing count is arguably an *effective*  $M_{\text{eff}} < N$  — the effective-number-of-independent-tests idea from genome-wide association studies (Nyholt 2004; Li & Ji 2005; Li, Yeung, Cherny & Sham 2012; Galwey 2009), where linkage disequilibrium between SNPs is the exact analogue of shared structure between strategies. Computing  $M_{\text{eff}}$  exactly from the participation ratio of the strategy correlation matrix,  $M_{\text{eff}} = N/(1+(N-1)\langle \rho^2 \rangle)$ , gives  $M_{\text{eff}} \approx 5$  on our pools ( $\langle \rho^2 \rangle \approx 0.2$ ). Plugging that into the corrected envelope manufactures  $\sim 500$  "survivors" on real markets — but they are spurious: at  $M_{\text{eff}} \approx 5$  the worst-of-3 envelope is  $\approx 0$ , so the haircut vanishes and raw Sharpes pass trivially. These "survivors" directly contradict the distribution-free bootstrap (§4.6), which makes no independence assumption and finds 0. The lesson, which we flag as a trap rather than a result: participation-ratio dimensionality is not the effective number of *tail* tests for a

maximum statistic, and naively importing the genomics effective-N heuristic into best-of-N Sharpe testing is anti-conservative — the bootstrap, not an analytic effective-N, must be the arbiter. (This is the kind of plausible cross-disciplinary import that would have produced a false discovery; we ran it, it broke, and we report the break.)

We tried a *second* analytic shortcut (`envelope_departure.py`): calibrate an empirical effective-N per generator by matching the realized best-of-N standardized in-sample Sharpe on a null market to the i.i.d.-Gaussian envelope  $E[\max \text{ of } N]$ . It also failed, and chasing *why* produced the most useful methodological result of this section. We first suspected serial correlation (the i.i.d. Sharpe  $SE \text{ var} = (1 + \frac{1}{2}SR^2)/T$  ignores it) and implemented Lo's (2002) serial-correlation-adjusted standard error with a Newey–West HAC factor (`theory.hac_variance_ratio, lo_sharpe_se`). The diagnostic (`lo_diagnostic.py`) is decisive: the HAC inflation is  $\eta^2 \approx 1.07$  (autocorrelation is negligible), yet the cross-sectional variance of the standardized Sharpe is  $\approx 4.6$ , not 1 — and Lo's correction barely moves it ( $4.6 \rightarrow 4.4$ ). So serial correlation is *not* the culprit. The real cause is structural heterogeneity: a generative search does not produce  $N$  noisy draws of a *single* null statistic; it produces  $N$  structurally *different* strategies (different drift exposure, different turnover-cost drag). A variance decomposition makes this exact —  $\sim 78\%$  of the cross-sectional variance of in-sample Sharpe is *between-strategy* (structural), only  $\sim 22\%$  is within-strategy estimation noise — so the best-of-N is dominated  $\sim 3.5:1$  by picking the most favorably-*structured* strategy, not the luckiest estimate. No standard-error or correlation correction can capture that — the multiple-testing " $N$  i.i.d. trials" premise fails on the *identical* part, not just the independence part. This is the deeper reason the distribution-free block bootstrap (§4.6), which resamples each strategy's actual return path and so preserves its structure, is load-bearing rather than a mere check. Three analytic shortcuts (participation ratio, envelope inversion, Lo HAC), three failures, one mechanism — and one tool that works.

## 5. Related work

- **Deflated Sharpe / false-strategy theorem.** Bailey & López de Prado, *The Deflated Sharpe Ratio* (2014) and *The Probability of Backtest Overfitting* (2016); López de Prado, *Advances in Financial Machine Learning* (2018). The DSR haircut is theirs; we apply it inside a multi-generator  $N$ -sweep with controls, and characterize its *power* (§4.5), which the original work does not.

- **Data-snooping tests we should be compared to (and partly do not yet run).** Sullivan, Timmermann & White (1999), "Data-Snooping, Technical Trading Rule Performance, and the Bootstrap" — *almost exactly our setup* (a rule universe  $\times$  snooping correction on an index), and the closest prior art; White (2000) Reality Check; Hansen (2005) Superior Predictive Ability; Romano & Wolf (2005) stepwise multiple testing; Harvey, Liu & Zhu (2016) and Harvey & Liu, "Backtesting" (2015) and "Lucky Factors" (2021) on the factor zoo and the  $t > 3$  threshold; Barras, Scaillet & Wermers (2010) on FDR in fund returns. We run the White/Romano–Wolf studentized Reality Check and the Hansen SPA as a distribution-free complement to the parametric DSR (§4.6) — they agree on the zero.
- **The closest direct precedent — FINSABER (Li et al., arXiv 2505.07078, KDD 2026).** Peer-reviewed; finds LLM trading agents (FinMem, FinAgent) generate no statistically significant alpha ( $p > 0.34$ ) under a broad, long-horizon, survivorship-corrected evaluation, and concludes "LLM-derived alpha is likely a methodological artefact of narrow, biased evaluations." Our headline conclusion — that LLM trading edge evaporates under rigorous evaluation — is therefore already established prior art. FINSABER differs in *apparatus*: it controls snooping by *evaluation design* (20-year horizon, delisted names, rolling windows) and runs no deflated-Sharpe, White RC/Hansen SPA, N-sweep, power surface, or planted-edge controls. We are the formal-multiple-testing + N-sweep + calibrated-control complement to FINSABER's design-based result.
- **Same-lineage LLM alpha mining.** AlphaAgent (KDD 2025; targets alpha *decay* via AST regularization), QuantaAlpha (2602.07085; evolutionary LLM mining for discovery), AlphaForgeBench (2602.18481; benchmarks frontier LLMs), Lopez-Lira & Tang (2023), FinGPT, BloombergGPT. Verified: none applies multiple-testing correction, planted/noise controls, or an N-sweep power surface — so our methodological apparatus is differentiated within this lineage, but none of them is the source of the "edge vanishes" conclusion (that is FINSABER).
- **Verified LLM discovery & quality-diversity.** FunSearch (*Nature* 2023), AlphaEvolve; MAP-Elites (Mouret & Clune 2015); novelty search (Lehman & Stanley 2011); Boden, *The Creative Mind* (operator taxonomy).
- **Worst-of-regimes scoring and maximum-Sharpe inference.** Scoring a strategy by its lowest risk-adjusted return across regimes is the minimum-regime-performance (MRP) criterion of Alexander & Fabozzi (2026), used there as a durability diagnostic; inference on the *maximum* in-sample Sharpe over a set of candidates is developed by Pav (2019). We build directly on both — our

contribution is the multiple-testing deflation of the MRP-style min-over-regimes best-of-N statistic (§4.9), which neither provides.

**What is new here.** The statistical machinery is established: the multiple-testing gate is White (2000) / STW (1999) / Bailey–López de Prado (2014); STW(1999) already reported a data-snooping-adjusted zero-survivor result on S&P 500 futures (Reality-Check Sharpe p-values  $\approx 0.99$ ), transaction-cost-aware and OOS, on parameter-optimized rule grids; and the "discovery threshold rises with N" curve is the leading-order asymptotic of BLdP Eq. 1, whose Exhibit 2 already plots the threshold rising with N (so our "power surface" is the *empirical instantiation* of a known result, not a new theorem). What the surveyed literature does not contain, and what we contribute, is narrow but real: (i) a multiple-testing deflation of the minimum-regime-performance statistic (Alexander & Fabozzi 2026) — the expected max of N of the min over k regimes,  $E[\max_n \min_k Z]$ , which is  $\sim 2.4\times$  smaller than the single-Sharpe DSR envelope the prior literature (including the maximum-Sharpe inference of Pav 2019) uses; deflating the *robustness* statistic correctly (rather than mis-applying the single-Sharpe haircut) is, to our knowledge, new, and it tightens the real-market margin from  $-1.5$  to  $-0.3$  while keeping the zero (§4.9); (ii) a cross-generator design — random vs evolutionary-creative vs a live LLM — through *one fixed gate*, isolating the generator's effect; (iii) planted-edge / pure-noise controls that *calibrate* the gate, absent from both the econometrics canon (one rule family) and the LLM-alpha work (real data only); (iv) a cautionary negative importing the GWAS effective-number-of-independent-tests machinery (Nyholt 2004; Li & Ji 2005; Galwey 2009) — it is anti-conservative for the best-of-N max statistic and the bootstrap must arbitrate (§4.9); and (v) the LLM diversity-ceiling finding (160 distinct from 1,000 draws). In one line: we re-confirm a known negative (STW; FINSABER) with a more formal, generator-controlled apparatus, and contribute a correct worst-of-regimes deflation, a cross-generator and planted-edge calibration, a cross-disciplinary cautionary negative, and the LLM diversity ceiling.

- **Effective number of independent tests (cross-disciplinary).** The correlated-tests problem we hit is solved in genome-wide association studies: Nyholt (2004), Li & Ji (2005), Li, Yeung, Cherny & Sham (2012), Galwey (2009) estimate the effective number of independent SNP tests from the correlation/LD eigenstructure; Carvajal-Rodríguez et al.'s SGoF even gains power as the number of tests grows. Finance's deflated-Sharpe/White-RC assume the full N; we import the genomics framing and

report (§4.9) that the naive participation-ratio version fails for the maximum statistic — a transfer that *looks* right and isn't.

## 6. Limitations (honest)

- **Low recall is the headline caveat, not a footnote (§4.5).** The gate's zero on real data is a *precision* statement; its power curve shows it misses any edge with a true worst-regime Sharpe below  $\approx 2-3$  at  $N=1,000$ . So "0 survivors" must always be read as "no edge provable at this search scale," never as "no edge exists." This is intrinsic to multiple-testing at scale, not a bug, but it bounds every claim in the paper.
- **The worst-regime statistic is not the statistic the DSR was derived for — *now addressed*.** Bailey–López de Prado's expected-max-under-null is for a single in-sample Sharpe over  $N$  trials; we apply the haircut to the *minimum across regimes*, whose null distribution is not standard-normal, so the DSR is conservative-but-heuristic here. §4.6 closes this with a distribution-free stationary block bootstrap (White Reality Check / Romano–Wolf / Hansen SPA) on the worst-regime statistic itself; it reaches the identical zero on real data. The DSR remains the primary gate (cheap, per-candidate); the bootstrap is the nonparametric check.
- **The DSL and data are modest.** Close-only FRED series disable the range/volume features; the feature set is small and daily; no cross-sectional, intraday, or alternative data. A richer DSL might surface a survivor — the method ( $N$ -sweep + ablation + power curve), not the specific null, is the transferable result.
- **Reported counts are finite-pool quantities, not independent-per- $N$  draws.** `bestOOS` and the winner/robust/survivor counts are bootstrap means over nested subsets of one fixed pool ( $\leq 4,000$  per market-generator); for  $N$  near the pool there is essentially one sample. They estimate within-pool expectations, not an independent accumulation across  $N$  — the monotone "rise with  $N$ " is a property of  $E[\text{best-of-}N]$ , reported as such.
- **Seed variance — now quantified (`seed_variance.py`).** Redrawing each pool under 8 independent seeds, the real-market survivor count is  $0 \pm 0$  under every seed at every  $N$  (`bestOOS` bands  $\pm 0.00-0.07$ ), and the planted control is positive under every seed for the creative generator at every  $N$  and for random at  $N \geq 300$  (creative  $N=1,000$ :  $63 \pm 7$ ) — the single-seed curves are representative, not lucky draws. Residual: daily Sharpes are  $\sqrt{252}$ -annualized with no serial-correlation adjustment, which can flatter the "annual Sharpe  $\sim 1.0$ " rhetoric.

- **Synthetic controls are stylized.** The strong planted edge is high-frequency mean reversion (deliberately easy, to verify the gate *can* pass); the noise market's adversarial regimes make worst-regime alone sufficient there, which is *why* the real-market ablation (where it is insufficient) is the informative case. The `small_planted` and power controls (§4.5) address "is the null rigged easy?" directly.
- **Backtests certify the past.** Even a survivor would be provisional; markets are adversarial and non-stationary.

## 7. Future work

Most of the v0.2 future-work list is now done (§4.4–§4.8). What remains:

- **Extend the LLM arm from N=300 to N=3,000.** The diversity ceiling (§4.4) suggests diminishing returns, but confirming the full curve with a live model is the last gap between the offline scaling story and the LLM one. The cached-pool mechanism makes it a cost question ( $\sim 10\times$  the current API budget), not an engineering one. A multi-LLM comparison (GPT, Gemini, an open model) would also test whether the diversity ceiling is Claude-specific.
- Romano–Wolf *step-down* (more powerful than the single-step we run) and extending the bootstrap to the panel and independent-market settings.
- **Richer DSL / intraday data.** A larger operator set and live-volume intraday bars, to test whether the zero is a property of the gate or of the signal space — and a monthly-rebalanced cross-sectional factor variant (lower turnover than the daily panel).
- **Position the finding as a general failure mode.** "Verified generative search under fixed effect size and growing N" also describes molecule screening, neural-architecture search, and LLM hypothesis generation; one discussion section would broaden the citation surface beyond finance.

## 8. Reproducibility

```
cd ~/mentat                                     # use python3.14
python3.14 -m mentat.n sweep --pool 4000 --data data/fred_SP500.csv \
--out papers/false-alpha/n sweep_results.json   # §4.1-
4.3
python3.14 papers/false-alpha/power.py          # power surface
§4.5 -> power_surface.json
python3.14 papers/false-alpha/bootstrap.py       # distribution-
```

```

free test $4.6 -> bootstrap_results.json
python3.14 papers/false-alpha/fetch_panel.py && \
python3.14 papers/false-alpha/panel_sweep.py                # cross-sectional
panel $4.7 -> panel_results.json
python3.14 papers/false-alpha/fetch_markets.py && \
python3.14 papers/false-alpha/independent_markets.py        # FX/gold/crypto
$4.8 -> independent_markets.json
python3.14 papers/false-alpha/verify.py                     # 20-check
adversarial suite (20 PASS / 0 FAIL)
python3.14 papers/false-alpha/plot.py                       # render figures
(needs matplotlib)
# live-LLM arm (needs anthropic + key sourced from ~/swechats/.env; cached after
first run):
set -a && . ~/swechats/.env; set +a
~/swechats/.venv/bin/python papers/false-alpha/task5_llm.py 1000 # $4.4 ->
task5_llm_results.json

```

Offline arms: no dependencies, no API key, deterministic (CRC-seeded; two runs byte-identical). The 74-test engine suite is `python3.14 -m tests.test_core`. Core symbols: deflated-Sharpe = `trade_lab.expected_max_sharpe_under_null;` gate = `AlphaProblem.verify;` cross-sectional portfolio = `panel_lab.panel_portfolio;` effective-trials = `nsweep.effective_trials;` `bootstrap = bootstrap.py`. Every reported number lives in a committed \*\_results.json. LLM pools are cached at `papers/false-alpha/llm_cache_*.json` so the live arm reproduces without re-billing the API.

## 9. Authorship and contributions

**Author:** Parv Mehndiratta — conceived the research question, designed the N-sweep and control structure, specified and built the experiment (`nsweep.py`) on the author's prior alpha engine (`trade_lab.py`, `imagine.py`), ran the analysis, and wrote the paper.

**Tools disclosed:** Implementation and drafting were done with AI assistance (Claude, Anthropic) used as a coding and writing aid under the author's direction; the underlying Mentat system uses a frozen LLM as a reasoning core, as disclosed in the Mentat paper. No undisclosed co-authors. (A previously referenced CRRA/Heston collaboration is *not* part of this work and is not claimed here pending independent verification.)

## 10. Reproducibility and disclosure

Every reported number is produced by a committed, CRC-seeded, deterministic script and written to a JSON artifact; two runs are byte-identical. The full engine, controls, and figures are public at [github.com/0103-parv/mentat](https://github.com/0103-parv/mentat), and correctness is checked by an executable suite (`verify.py`, 20/20) and 38 math/invariant unit tests (`test_theory.py`). Implementation and drafting used AI assistance (Claude, Anthropic) under the author's direction, as disclosed in §9; the underlying Mentat system uses a frozen LLM as a reasoning core.

The draft was hardened through several adversarial review passes — statistical, positioning, and reproducibility — before release. These overturned two of the author's own early claims, reported here in that spirit: the "creativity snoops more" hypothesis was deleted once the mean correlation came in at  $\rho \approx 0$ , and the "power surface" was reccredited as the empirical instantiation of Bailey–López de Prado's expected-max result rather than a new theorem (§4.5). The decisive critique — that the positive control was too easy and the null possibly rigged — produced the paper's central power-curve result (§4.5), reframing the contribution from "no edge in the S&P" to the precision/recall statement that moderate edges are unprovable at scale. Residual honest gaps are listed in §6–§7.

## Acknowledgments

The author thanks Antoine Cully, who pointed him to the Extract-QD line of work on the winner's curse in quality-diversity search; Joel Lehman and Jean-Baptiste Mouret for helpful correspondence on selection inflation, novelty search, and evaluator overfitting; Robert Lange for discussion of open-ended LLM program search; and John Yang for discussion of held-out and chronological evaluation of coding agents. These exchanges shaped how the evaluation in this paper is framed. No acknowledgment implies endorsement of this work; all views and any errors are the author's own.

## References

*Classical and statistical references below are full bibliographic entries. The recent LLM-alpha-mining lineage discussed in §5 (AlphaAgent, QuantaAlpha, AlphaForgeBench, AutoQuant) is cited in-text with the identifiers used there; verify each arXiv id against arXiv before final submission.*



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## Appendix A — key numbers at a glance

- **Real S&P 500**, all 5,994 tradeable alphas (random + creative): best worst-regime annualized Sharpe = +0.847; best naive pooled-OOS = +0.999; survivors = 0 at

every N, all three generators (random, creative, live LLM).

- **Deflation haircut** on real S&P at  $n_{\text{trials}} = \{1000, 3000\} = \{2.31, 2.52\}$  ann.  $\rightarrow$  best deflated =  $\{-1.46, -1.67\}$ . Even at the smallest defensible *effective* trial count ( $\sim 34$ , random's  $N_{\text{eff}}$  at  $N=1,000$ ), the haircut ( $\sim 1.5$ ) still exceeds  $+0.847 \rightarrow$  deflated  $< 0 \rightarrow 0$  survivors is not an over-deflation artifact.
- **Search redundancy**  $\bar{\rho}$  on real S&P: random  $+0.029$ , creative  $-0.050$ , live LLM  $+0.007$  (mean correlation); but  $\langle \rho^2 \rangle \approx 0.2$  and no single analytic effective-N is reliable (§4.9) — the bootstrap is the arbiter.
- **Worst-of-regimes deflation (§4.9)**: standard DSR over-deflates the min-over-k statistic  $\sim 2.4\times$ ; corrected bar still gives 0 on real (best deflated  $\approx -0.3$ ). Robustness dividend: worst-of-3 at  $N=1,000 \approx$  single-Sharpe search over  $N' \approx 7$  (independent regimes) — robustness *is* deflation, collapsing to  $N' \approx 84$  on real S&P ( $\rho_{\text{reg}} \approx 0.67$ ) when regimes correlate.
- **Gate power curve** ( $N=1,000$ ): admits 0 survivors until the true worst-regime Sharpe exceeds  $\sim +2$  (edge giving Sharpe  $+1.94 \rightarrow 0$  survivors; only the strong edge at  $+4.47 \rightarrow 61$ ). A realistically small edge (`small_planted`, naive OOS  $\approx +0.5$ ) is invisible at every N — like the real S&P. The discovery bar grows  $\approx \sqrt{2 \ln N}$ .
- **Live LLM (Claude Opus 4.8)**: H1 ✓ (bestOOS  $+0.43 \rightarrow +0.79$  on real), H2 ✗ refuted ( $\bar{\rho} = +0.007$ , not more correlated than random), H3 ✓ (0 survivors on real), H4 ✓ (recovers strong planted edge faster than random).
- **Strong planted control**: survivors grow with N (creative  $0.7 \rightarrow 154$  at  $N=3,000$ ) — the gate passes a strong real edge.
- **Creativity/LLM on real S&P**:  $\sim 2\times$  the regime-robust mirages of random (196 vs 115), 0 survivors regardless.
- **Generalization (Tasks 2 & 4)**: the zero holds on a real 15-stock cross-sectional panel (range/volume active, 0 survivors all N) and on three microstructurally independent real markets — FX (best naive  $+0.2$ ), gold ( $+0.9$ ), crypto ( $+0.9$ ) — all 0 survivors at every N. Seven real markets, zero survivors.
- **Distribution-free check (Task 1)**: stationary block-bootstrap White RC / Romano–Wolf / Hansen SPA on the worst-regime statistic  $\rightarrow 0$  survivors on real S&P at every N, all 3 generators (RC p 0.15–0.72), stable across block lengths  $\{10, 25, 50\}$ ; grows with N on strong planted. The zero is not a parametric-DSR artifact.
- **Power surface (Task 3)**: detectable-edge threshold rises with N — edge 0.25 admitted at  $N=10$  but rejected at  $N \geq 30$ ; only the 0.40 edge stays admitted ( $2 \rightarrow 168$

survivors as  $N$  goes  $10 \rightarrow 3000$ ).

- **Adversarial verification:** `verify.py`  $\rightarrow$  PASS / 0 FAIL ( $\text{cost} \in \{0, \dots, 0.002\}$ ,  $\text{target} \in \{0, 0.25, 0.5\}$ , 2/3/5 regimes; S&P, DJIA, NASDAQ, FX, gold, crypto all 0 survivors; independent `curriculum.study` agrees 0/6 verified facets; positive control passes the real `AlphaProblem.verify`, engine score +1.06).